In [1]:
```
import numpy as np
import pandas as pd

list = [[1, 'Smith', 50000],
        [2, 'Jones', 60000]]

df = pd.DataFrame(list)
df
```

Out[1]:

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

In [2]:
```
df.columns = ['Empd', 'Name', 'Salary']
df
```

Out[2]:

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

In [3]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      str
 2   Salary  2 non-null      int64
dtypes: int64(2), str(1)
memory usage: 180.0 bytes
```

In [6]:
```
import pandas as pd

data = {
    'R&D Spend': [165349.20, 162597.70, 153441.51, 144372.41, 142107.34],
    'Administration': [136897.80, 151377.59, 101145.55, 118671.85, 91391.77],
    'Marketing Spend': [471784.10, 443898.53, 407934.54, 383199.62, 366168.42],
    'State': ['New York', 'California', 'Florida', 'New York', 'Florida'],
    'Profit': [192261.83, 191792.06, 191050.39, 182901.99, 166187.94]
}

df = pd.DataFrame(data)

df
```

Out[6]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

In [7]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   R&D Spend        5 non-null      float64
 1   Administration   5 non-null      float64
 2   Marketing Spend  5 non-null      float64
 3   State            5 non-null      str    
 4   Profit           5 non-null      float64
dtypes: float64(4), str(1)
memory usage: 332.0 bytes
```

In [8]:
```
df.head()
```

Out[8]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

In [9]:
```
df.tail()
```

Out[9]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

In [10]:
```
df.describe()
```

Out[10]:

	R&D Spend	Administration	Marketing Spend	Profit
count	5.000000	5.000000	5.000000	5.00000
mean	153573.632000	119896.912000	414597.042000	184838.84200
std	10443.342374	24727.114441	43314.129541	11108.99254
min	142107.340000	91391.770000	366168.420000	166187.94000
25%	144372.410000	101145.550000	383199.620000	182901.99000
50%	153441.510000	118671.850000	407934.540000	191050.39000
75%	162597.700000	136897.800000	443898.530000	191792.06000
max	165349.200000	151377.590000	471784.100000	192261.83000

In [11]:
```
df.mean(numeric_only=True)
```

Out[11]:
```
R&D Spend          153573.632
Administration     119896.912
Marketing Spend    414597.042
Profit             184838.842
dtype: float64
```

In [12]:
```
df.median(numeric_only=True)
```

Out[12]:
```
R&D Spend          153441.51
Administration     118671.85
Marketing Spend    407934.54
Profit             191050.39
dtype: float64
```

In [13]:
```
df.mode()
```

Out[13]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	142107.34	91391.77	366168.42	Florida	166187.94
1	144372.41	101145.55	383199.62	New York	182901.99
2	153441.51	118671.85	407934.54	NaN	191050.39
3	162597.70	136897.80	443898.53	NaN	191792.06
4	165349.20	151377.59	471784.10	NaN	192261.83

In [14]:
```
df.var(numeric_only=True)
```

Out[14]:
```
R&D Spend          1.090634e+08
Administration     6.114302e+08
Marketing Spend    1.876114e+09
Profit             1.234097e+08
dtype: float64
```

In [15]:
```
df.std(numeric_only=True)
```

Out[15]:
```
R&D Spend          10443.342374
Administration     24727.114441
Marketing Spend    43314.129541
Profit             11108.992540
dtype: float64
```

In [16]:
```
df.describe()
```

Out[16]:

	R&D Spend	Administration	Marketing Spend	Profit
count	5.000000	5.000000	5.000000	5.00000
mean	153573.632000	119896.912000	414597.042000	184838.84200
std	10443.342374	24727.114441	43314.129541	11108.99254
min	142107.340000	91391.770000	366168.420000	166187.94000
25%	144372.410000	101145.550000	383199.620000	182901.99000
50%	153441.510000	118671.850000	407934.540000	191050.39000
75%	162597.700000	136897.800000	443898.530000	191792.06000
max	165349.200000	151377.590000	471784.100000	192261.83000

In [17]:
```
df.describe(include='all')
```

Out[17]:

	R&D Spend	Administration	Marketing Spend	State	Profit
count	5.000000	5.000000	5.000000	5	5.00000
unique	NaN	NaN	NaN	3	NaN
top	NaN	NaN	NaN	New York	NaN
freq	NaN	NaN	NaN	2	NaN
mean	153573.632000	119896.912000	414597.042000	NaN	184838.84200
std	10443.342374	24727.114441	43314.129541	NaN	11108.99254
min	142107.340000	91391.770000	366168.420000	NaN	166187.94000
25%	144372.410000	101145.550000	383199.620000	NaN	182901.99000
50%	153441.510000	118671.850000	407934.540000	NaN	191050.39000
75%	162597.700000	136897.800000	443898.530000	NaN	191792.06000
max	165349.200000	151377.590000	471784.100000	NaN	192261.83000

In [18]:
```
empCol = df.columns
empCol
```

Out[18]:
```
Index(['R&D Spend', 'Administration', 'Marketing Spend', 'State', 'Profit'], dtype
='str')
```

In [19]:
```
emparray = df.values
emparray
```

Out[19]:
```
array([[165349.2, 136897.8, 471784.1, 'New York', 192261.83],
       [162597.7, 151377.59, 443898.53, 'California', 191792.06],
       [153441.51, 101145.55, 407934.54, 'Florida', 191050.39],
       [144372.41, 118671.85, 383199.62, 'New York', 182901.99],
       [142107.34, 91391.77, 366168.42, 'Florida', 166187.94]],
      dtype=object)
```

In [20]:
```
employee_DF = pd.DataFrame(emparray, columns=empCol)
employee_DF
```

Out[20]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.2	136897.8	471784.1	New York	192261.83
1	162597.7	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

In [ ]: